

usb_joystick_routed

Design Report

Rev 1 | 2026-09-14 | jlcpcb-tier1

kicad-tools 0.20.0

Release Status: BLOCKED

This candidate is not qualified for manufacture. The saved copper has 12 native copper-silver warnings and does not match independent bare native refill. The passing recipe replay and factory checks below do not clear these release blockers. See Manufacturing Readiness and the bundled `native-refill-comparison.json` for the measured discrepancy.

This report's release evidence was corrected against PR #5386 source `aa1fa691f545866a7754e28e1baf773e8cae31a8` . Measurements are retained KiCad 10.0.5 observations, not a new physical run. Board imagery, geometry and recipe remain unchanged.

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	38 (37 SMD, 1 THT, 0 other)
Nets	27
Traces	2050 segments
Vias	153
Board Size	80.0 x 60.0 mm

Stackup

Layer	Type	Thickness (mm)	Material
F.Cu	copper	0.035	--
dielectric 1	prepreg	0.2104	7628
In1.Cu	copper	0.0152	--
dielectric 2	core	1.065	FR4
In2.Cu	copper	0.0152	--
dielectric 3	prepreg	0.2104	7628
B.Cu	copper	0.035	--

Design Overview

Theory of Operation

USB joystick controller — ATmega32U4

USB bus powered, 8MHz crystal, AVR ISP programming

GPIO and ADC map is fixed to Microchip TQFP-44 pinout

Communication Interfaces

Protocol	Signals
SPI	ISP_MISO, ISP_MOSI, ISP_SCK
USB	USB_D+, USB_D-, VBUS

Assembly Notes

1 fine-pitch component

- **Fine-pitch components:** 1 (U1)

Hand-Solder (THT) Components

The following 1 through-hole component is **excluded from the SMT pick-and-place file** and must be hand-soldered (or wave/selective-soldered) after SMT assembly. It appears in the BOM for sourcing.

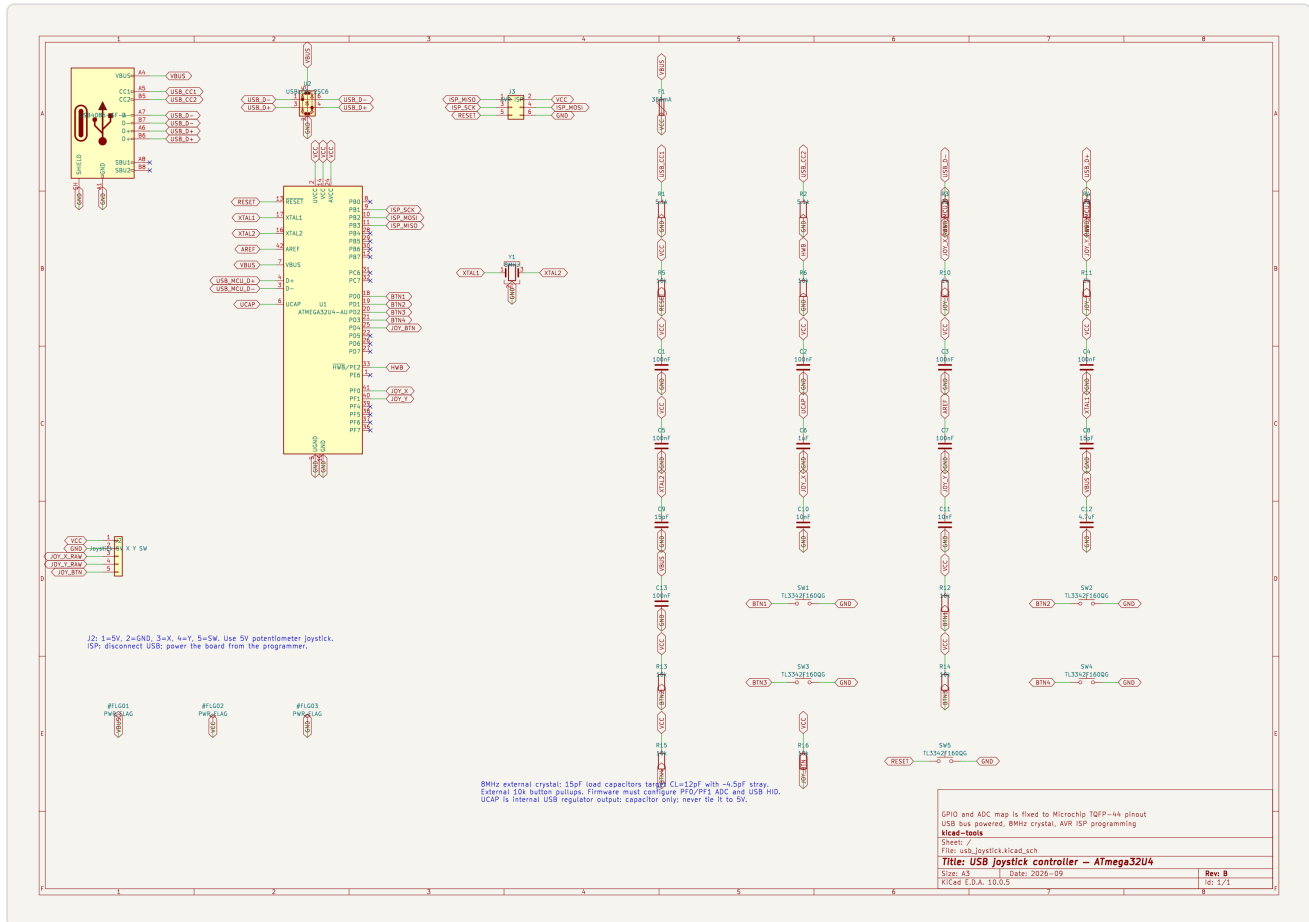
Value	Package	Qty	References
USB4085-GF-A	USB_C_Receptacle_GC T_USB4085	1	J1

ERC Status

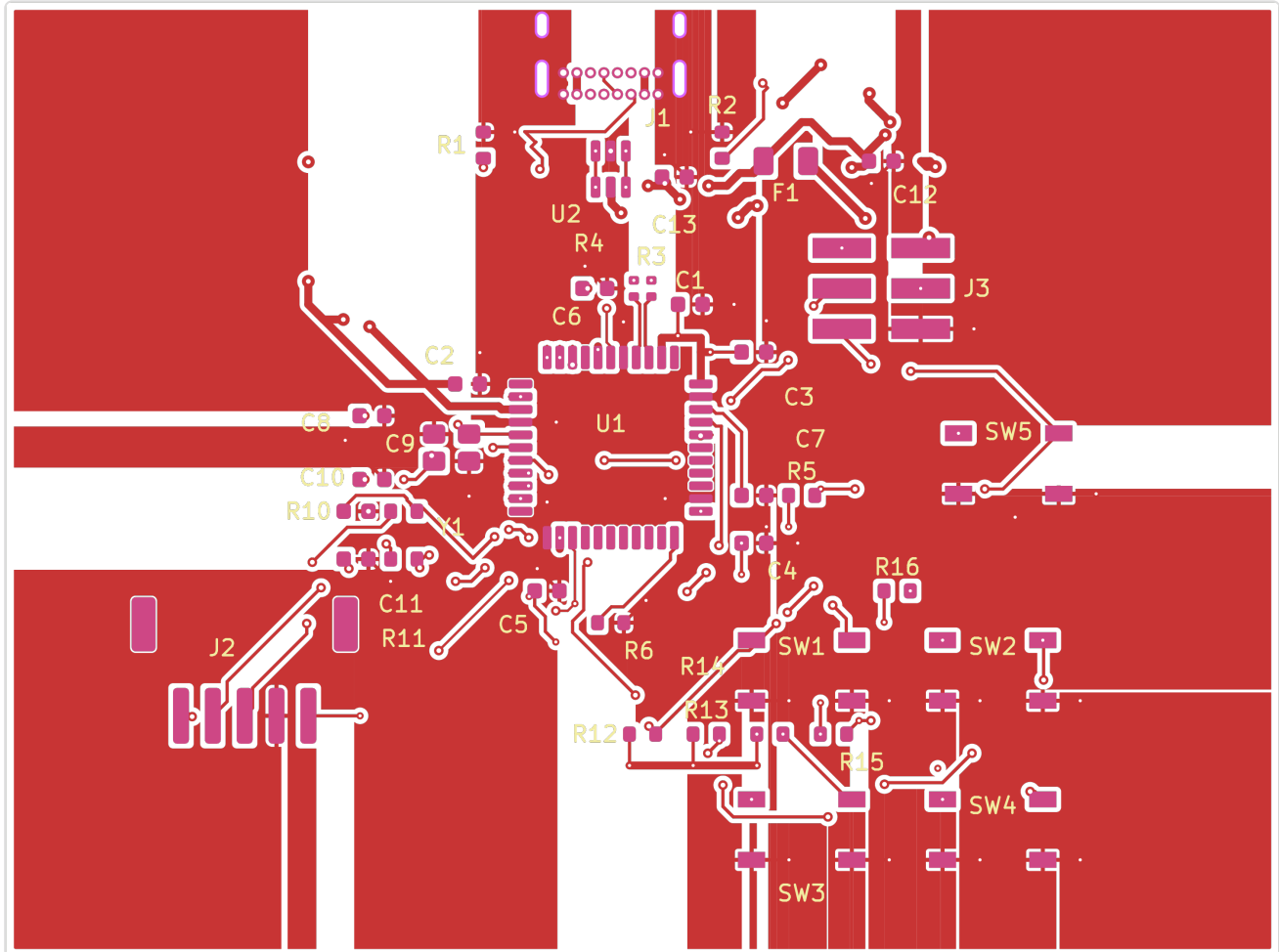
Retained native KiCad 10.0.5 report dated 2026-09-15T00:37:00 on the shipped schematic: **0 errors, 0 warnings**. Evidence: native-erc.json ; schematic UUID /546c89fe-5075-48a6-a97e-bef2a1207261 . This supersedes the earlier export-stage ERC-skipped summary. No new ERC run was performed for this report correction.

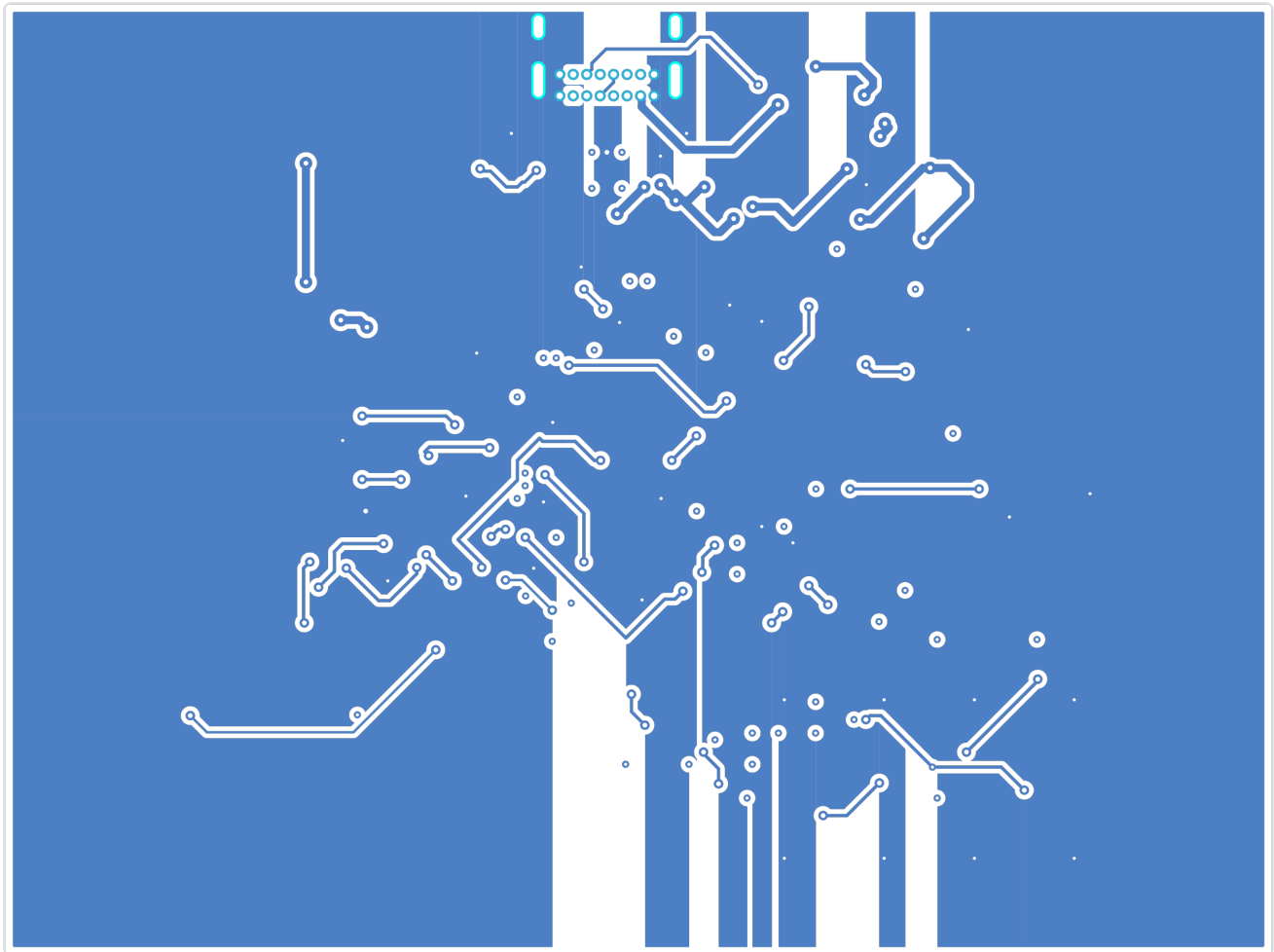
Schematic Overview

Schematic: usb_joystick

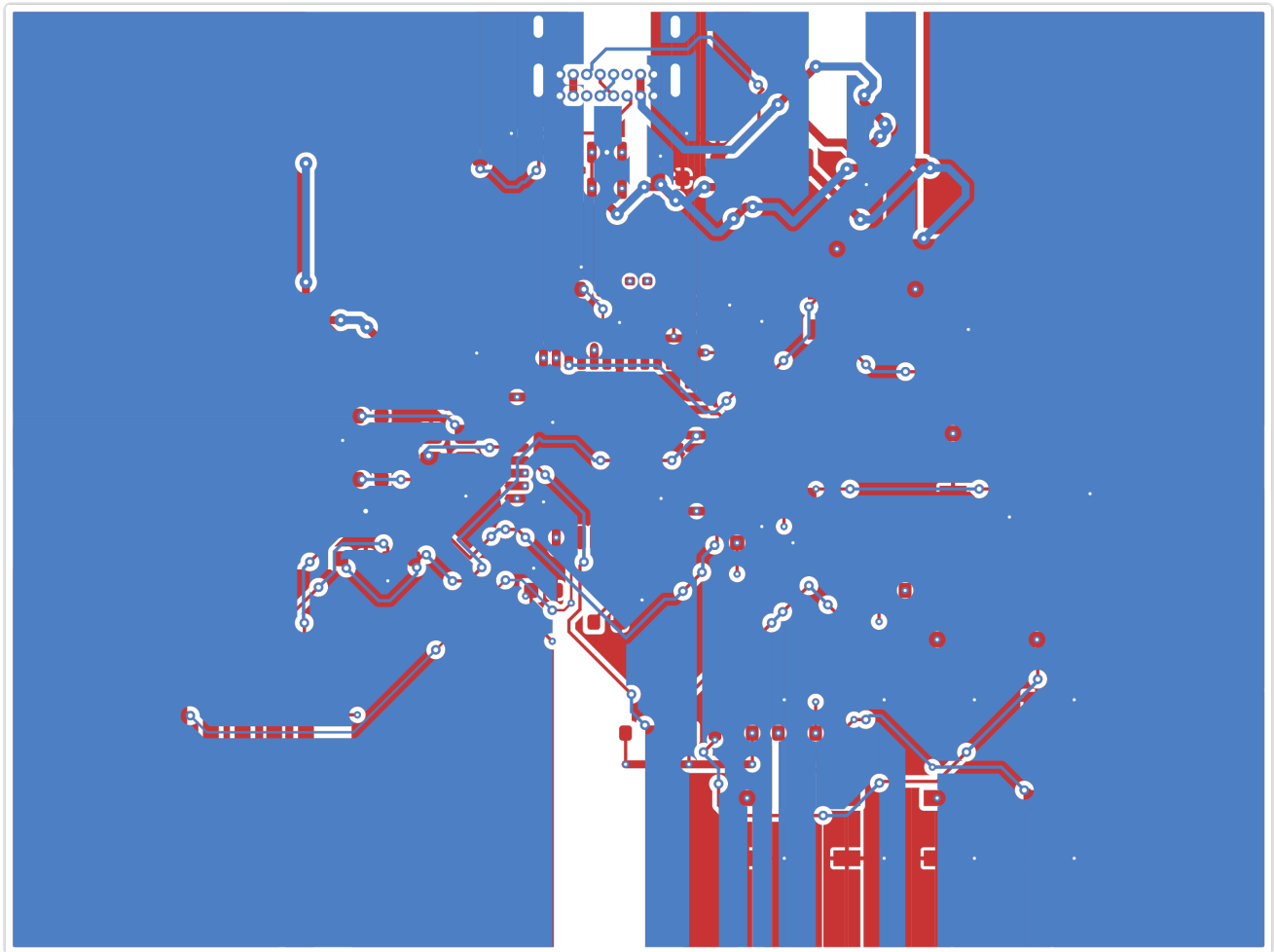


PCB Layout

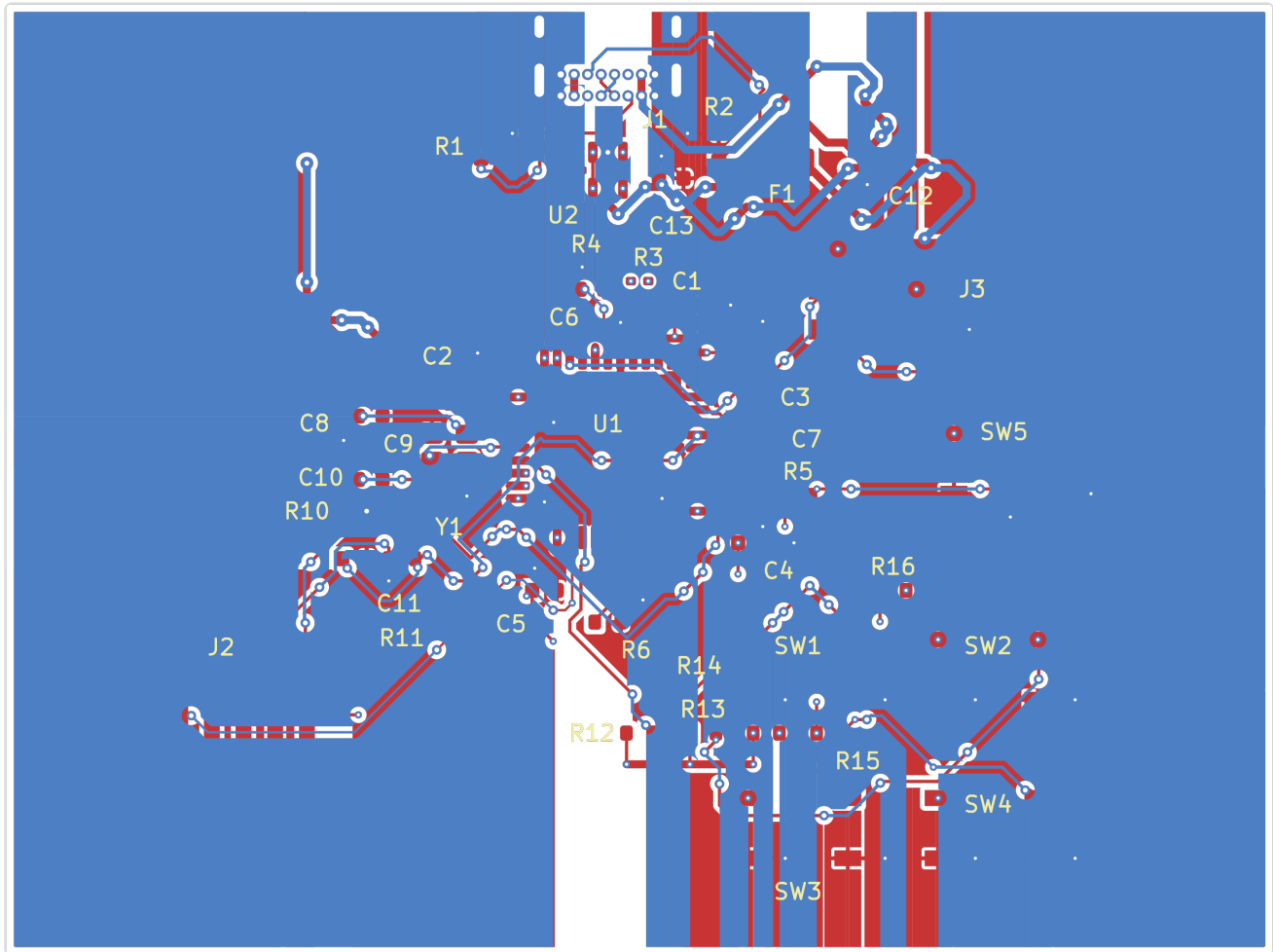




Copper

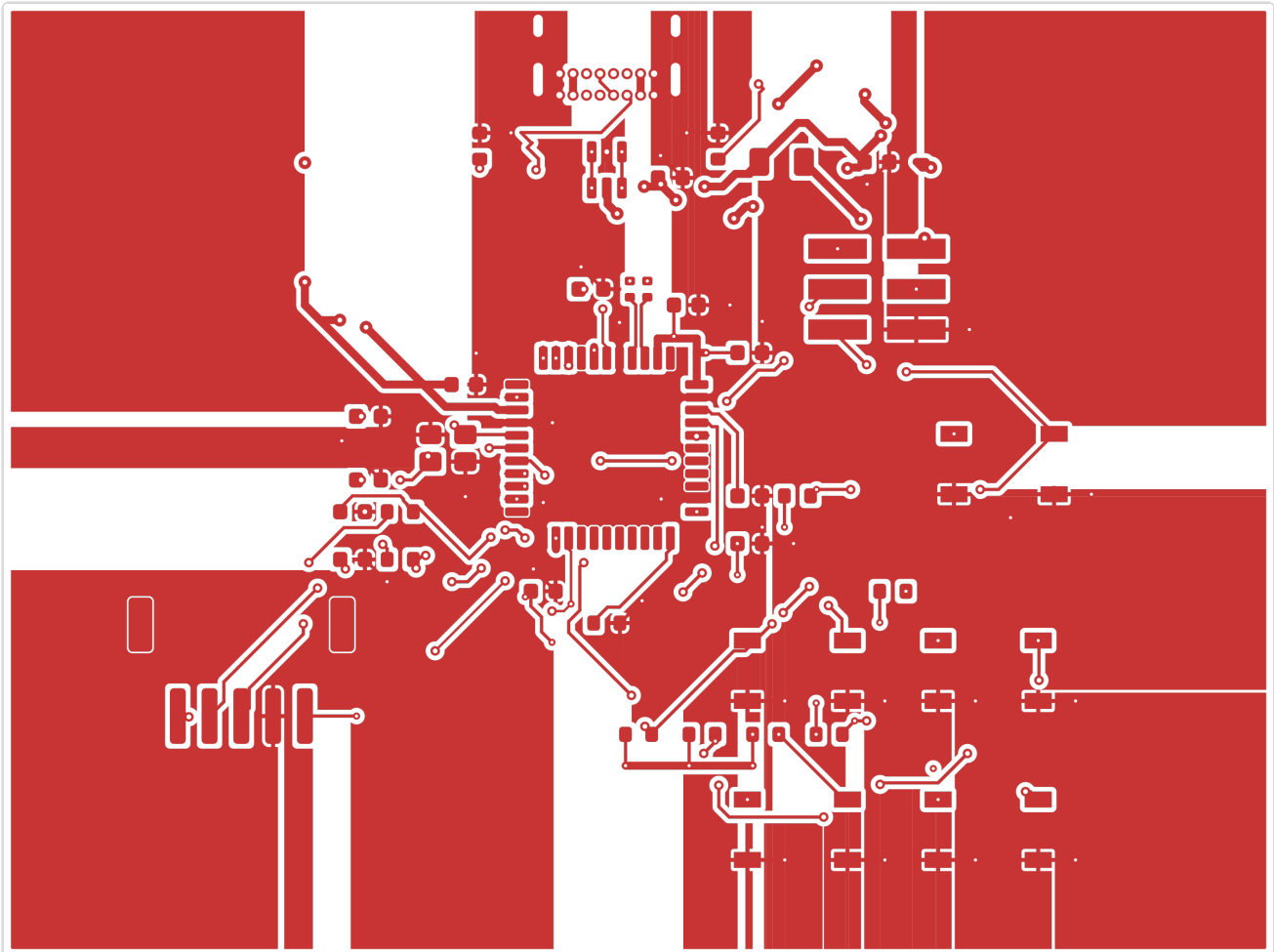


Assembly

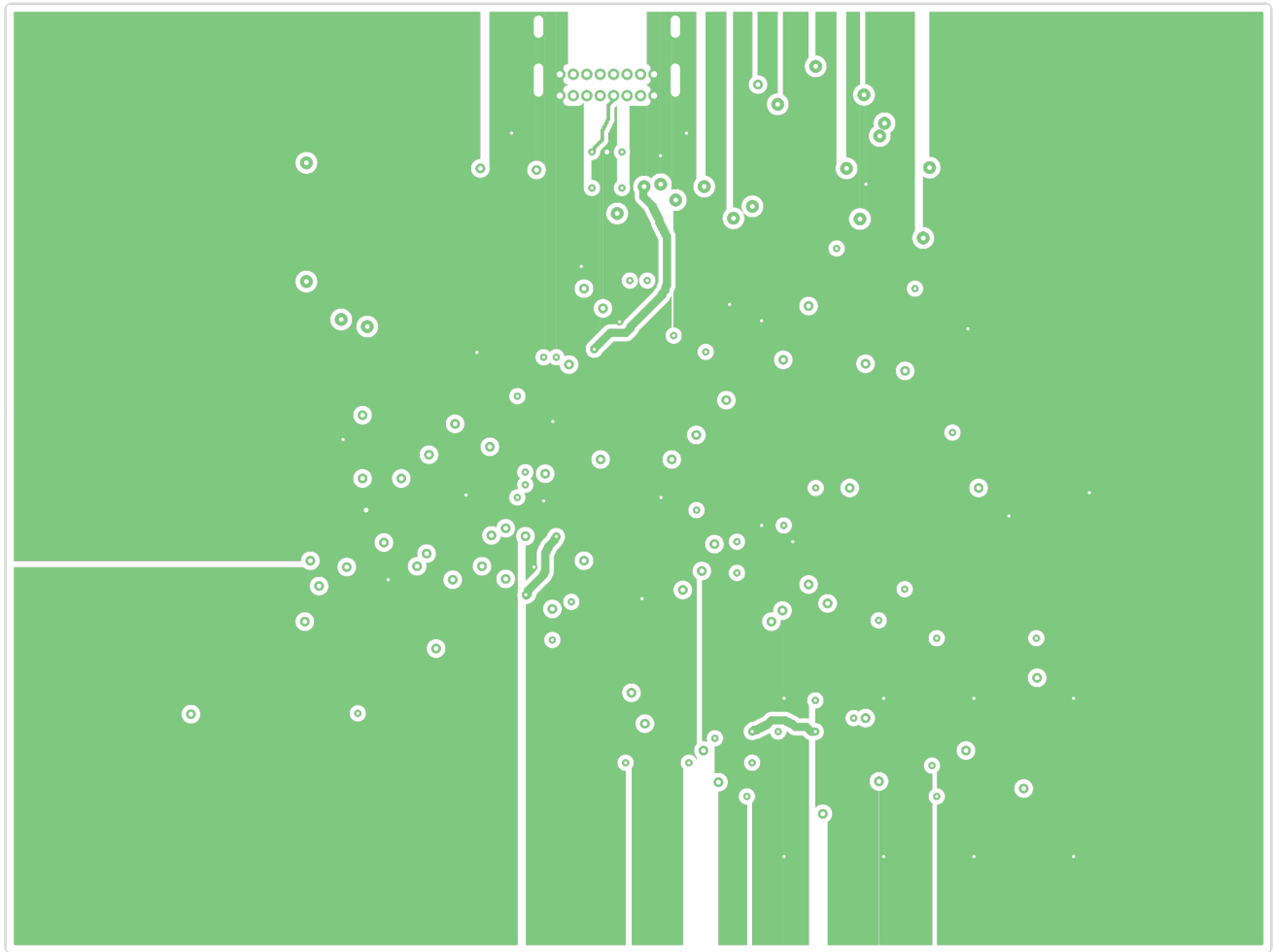


Copper Layers

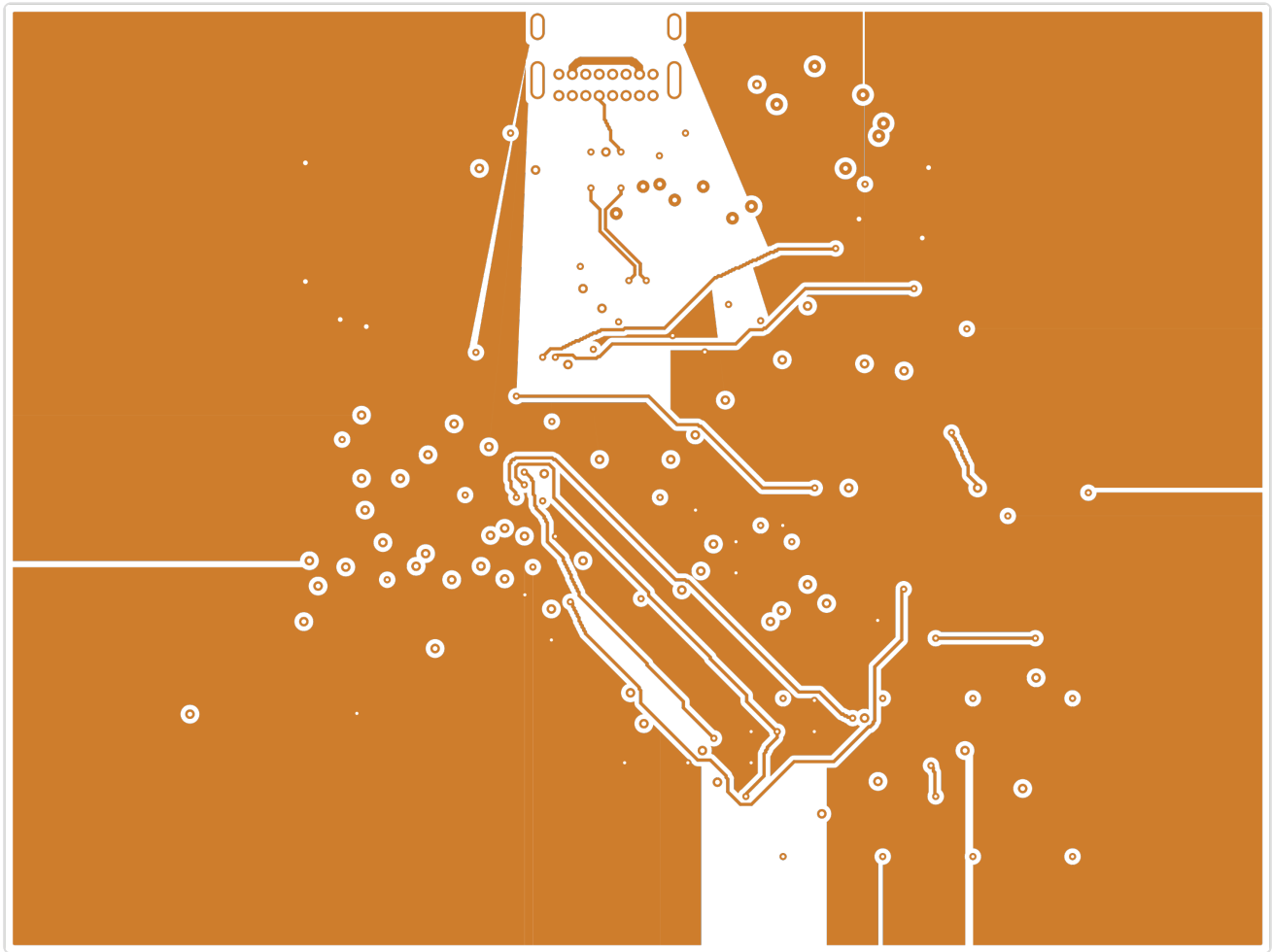
F.Cu



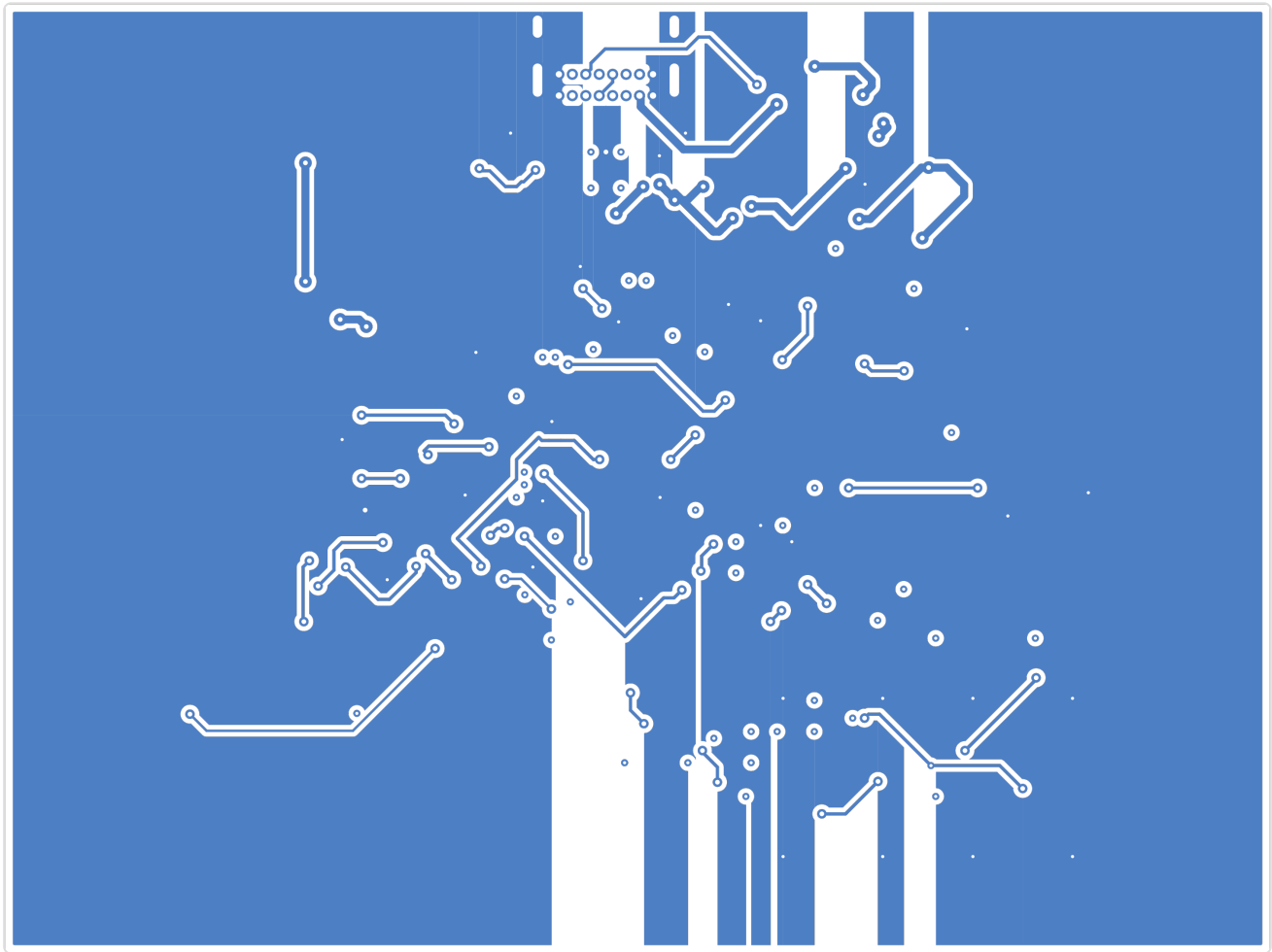
In1.Cu



In2.Cu



B.Cu



Bill of Materials

Value	Package	Qty	References	MPN	LCSC
100nF	C_0603_1608 Metric	7	C1, C2, C3, C4, C5, C7, C13	GRM188R71C 104KA01D	C45000
10nF	C_0603_1608 Metric	2	C10, C11	GRM188R71H 103KA01D	C77053
15pF	C_0603_1608 Metric	2	C8, C9	GRM1885C1H 150JA01D	C71651
1uF	C_0603_1608 Metric	1	C6	GRM188R71A 105KA61D	C97888
4.7uF	C_0603_1608 Metric	1	C12	GRM188R61A 475KE15D	C71633
350mA	Fuse_1206_32 16Metric	1	F1	1206L035/16Y R	C126817
AVR ISP	Samtec_TSM- 103-01-T-DV	1	J3	TSM-103-01-T- DV-P-TR	C21285200
Joystick 5V X Y SW	JST_PH_S5B- PH-SM4- TB_1x05-1MP _P2.00mm_Ho rizontal	1	J2	S5B-PH-SM4- TB(LF)(SN)	C265104
USB4085-GF- A	USB_C_Recep tacle_GCT_US B4085	1	J1	USB4085-GF- A	C7095263
10k	R_0603_1608 Metric	7	R5, R6, R12, R13, R14, R15, R16	RC0603FR-07 10KL	C98220
1k	R_0603_1608 Metric	2	R10, R11	RC0603FR-07 1KL	C22548
22	R_0402_1005 Metric	2	R3, R4	RC0402FR-07 22RL	C114765
5.1k	R_0603_1608 Metric	2	R1, R2	RC0603FR-07 5K1L	C105580
TL3342F160Q G	SW_SPST_TL 3342	5	SW1, SW2, SW3, SW4, SW5	TL3342F160Q G	C2886898

Value	Package	Qty	References	MPN	LCSC
ATMEGA32U4-AU	TQFP-44_10x10mm_P0.8mm	1	U1	ATMEGA32U4-AU	C44854
USBLC6-2SC6	SOT-23-6	1	U2	USBLC6-2SC6	C7519
8MHz	Crystal_SMD_3225-4Pin_3.2x2.5mm	1	Y1	ECS-80-12-33-JGN-TR	C2442665

DRC Status: Saved Copper

Metric	Count
Native errors	0
Native warnings	12
Unconnected items	0
Copper-sliver warnings	12

Native release gate: FAILED. The zero-error/zero-warning gate is not satisfied. Evidence: bundled `native-drc.json`, measured on the canonical saved board with full project, rules and custom footprint context, without refill.

The separately scoped retained Python factory check in `check-report.json` passed (0 errors/warnings, 54 checked rules). Its pass does not replace native readiness checks. The former 14 hole-to-hole warning summary was an earlier export-stage diagnostic; it is superseded here and is not the current native warning census.

Native version: KiCad 10.0.5. Container image ID:

`sha256:182c8005cb775a2c448a4c18681d489f1ff472a761885eba3e08b07e3c0564de`.

Manufacturing Readiness

Verdict: **BLOCKED**

Required Actions

- **[REQUIRED]** Resolve the same-copper compatibility failure between the preserved recipe-filled release and bare native refill. Do not discard the clearance carve or substitute different refilled copper to claim success.
- **[REQUIRED]** Resolve the 12 native copper-sliver warnings on the actual shipped geometry and rerun the required zero-error/zero-warning gates.
- **[REQUIRED]** Verify the final candidate and its release bindings after repair. The generic producer correction in #5391 alone does not resolve this physical qualification gap; #5380 remains incomplete.

Independent Native-Fill Comparison

Layer	Refilled minus saved copper area (mm ²)
F.Cu	947.873802
B.Cu	406.966144
In1.Cu	217.304656
In2.Cu	463.102399

Saved PCB SHA256:

e7bdf5b80a3685f4ebd7e380fdade7d9e6218168a15f0cfecea6166b69645270 .

Separate native-refilled PCB SHA256:

81de0e491ad82a68cfe375737dbf898969d45e35febb5a2737649e2378bd8e5a .

The refilled measurement has zero native violations/opens but describes different copper and is not the shipped board. Evidence: `native-refill-comparison.json` and `native-refilled-drc.json` .

The retained recipe-fill replay reports zero area delta for its own native-fill/remediation/carve engine. That passing recipe-stability result is distinct from bare-native equivalence. Neither it nor passing factory/connectivity checks establishes overall release readiness.

Routing Status

Metric	Value
Signal Net Completion	100.0% (25/25)
Overall Completion	100.0%
Complete Nets	27 / 27
Zone-Connected Nets	2
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- GND
- VCC

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~2.96 USD
Components (estimated)	~2.11 USD
Assembly (estimated)	~2.16 USD
Total (estimated)	~7.23 USD
Batch Quantity	5
Batch Total (estimated)	~36.14 USD